DE RUEDA CONSTRUCTION ERP

Upgrades.pdf — Consolidated Upgrade Requirements

Purpose and implementation boundary

Upgrade the existing ERP in place. Preserve working modules, employee records, attendance, payroll history, deductions, projects, sites, reports, and historical payment evidence. This document is a requirements specification; it does not prove that the changes have been implemented, tested, or deployed.

1. CEO Vault Master Transaction Cut-Off

Requirement: Add a prominent, CEO-only master ON/OFF switch inside the secured CEO Vault. When OFF, the system must prevent new GCash and PayMongo transaction execution across every applicable site, payout workflow, and user interface. Transactions may be initiated only when the master switch is ON **and** the specific provider, account, user, site scope, approval, and transaction rules all permit it.

Required behaviour

• **Default and fail-safe:** initialize to OFF when first configured, and fail closed if the switch state cannot be safely read.

• **Global scope:** apply the same authoritative control to all GCash and PayMongo transaction paths, including API endpoints, queued jobs, batch payouts, retries, scheduled work, and manual release screens.

• **Server-side enforcement:** check the master switch immediately before a transaction is accepted or dispatched. Hiding or disabling front-end buttons alone is insufficient.

• **Immediate cut-off:** OFF blocks new transactions and stops queued, not-yet-submitted transactions from being dispatched. Already submitted or in-flight provider transactions cannot always be recalled; mark and reconcile their actual status instead of falsely reporting cancellation.

• **Safe re-enable:** turning ON requires CEO authentication and explicit confirmation. It removes only the global lock; it must not bypass provider credentials, role permissions, site scope, approval thresholds, account status, or other controls.

• **Audit trail:** record CEO identity, timestamp, prior/new state, reason, and result for every toggle. Make records tamper-resistant and visible to authorized auditors.

• **Resilience:** if the database/control service is unavailable, block new transaction execution. Use concurrency-safe state reads and prevent stale cached ON values from allowing a payout.

• **Visibility:** display current state, last changed by/time, and any transactions already in flight. Show clear blocked/rejected feedback and do not fabricate success receipts or references.

• **Recovery:** after turning OFF, reconcile any provider-submitted transactions and queue items before re-enabling where appropriate.

Important distinction

The master switch is an execution lock, not a payment reversal feature. A provider may already have accepted a transaction before the cut-off takes effect. The ERP must preserve the provider's real status, avoid duplicate retries, and use reconciliation or provider-supported cancellation/refund processes where available.

2. Wallet-style weekly payroll

Provide a wallet-style payroll dashboard with Total Payroll, Ready for Release, Awaiting Confirmation, Processing, Confirmed, Failed, and Needs Review. Default to a weekly site-wide payroll batch while allowing authorized individual employee payouts. Every employee payment must retain its own status, evidence, and transaction reference.

3. Role and site-based permissions

• **CEO:** oversight, master cut-off switch, and explicitly delegated approvals/distribution authority.

• **HR/Payroll Admin:** prepare and review payroll, employee details, deductions, and exceptions according to granted permissions.

• **Authorized Payroll Distributor:** execute permitted releases only when specifically authorized.

• **Site Supervisor:** attendance, site records, and exceptions; no payout access by default.

• **Read-only user:** view only; no transaction execution.

• Enforce permissions on the server, scope access by assigned site, record delegation and revocation, and keep an audit trail. Job title alone must not grant transaction authority.

4. GCash and PayMongo integration safeguards

• Use only approved official provider integrations and credentials. A setup dialog or linked account alone does not prove that payout/disbursement capability is enabled.

• Treat GCash and PayMongo as separate provider adapters beneath the CEO master cut-off. The global OFF switch blocks both; provider-specific availability and permissions still apply when ON.

• Use provider-confirmed status, webhooks where available, idempotency protection, reconciliation, and duplicate-payment safeguards.

• Never collect or store GCash MPIN/PIN or OTP in the ERP. Never use fake balances, mock success, fabricated transaction IDs, or invented receipts in production.

• Keep client PayMongo payments on hold unless separately approved for implementation.

5. Payment evidence and SMS payslips

• For cash payroll, require a clear photo of the employee receiving cash plus acknowledgement/evidence, then review and verify before marking the payment confirmed.

• For electronic payments, use provider status and reconciliation where available; screenshots may support review but do not by themselves guarantee settlement.

• Send the SMS payslip only after the payment is verified as successful. Include pay period, gross pay, itemized deductions, net pay, amount paid, transaction reference where applicable, and payout status.

• Keep evidence access restricted, log review decisions, and retain evidence according to the company's retention policy.

6. Build, testing, and release requirements

• Standardize on npm: keep package.json and package-lock.json synchronized, remove conflicting Bun lockfiles if present, run npm ci, then npm run build.

• Test OFF blocks every GCash and PayMongo transaction path, including API calls, batch releases, queued jobs, scheduled jobs, and retries.

• Test concurrent requests and a switch change during processing; prove that stale state cannot authorize a new transaction after OFF is effective.

• Test that ON does not bypass user permissions, provider status, site scope, approval rules, or missing credentials.

• Test audit records, in-flight reconciliation, idempotency, SMS timing, failure handling, and recovery.

• For Cloudflare Pages, verify the configured build output (dist/ if applicable), deploy only after successful tests, and verify the actual deployed version before declaring completion.

Acceptance criteria

Check	Pass condition
CEO control	Only an authenticated, authorized CEO can change the master switch.
OFF state	New GCash and PayMongo transactions are blocked across all server-side paths.
ON state	Transactions proceed only if all independent permissions and provider checks pass.
In-flight items	Already submitted transactions retain accurate provider status and are reconciled.
Auditability	Every switch change and blocked attempt is recorded with actor and timestamp.
No false success	No payment is reported successful without valid evidence/provider confirmation.
Release quality	Build, security tests, regression tests, and live deployment verification pass.

Release rule: Do not claim the CEO Vault cut-off is live until the server-side enforcement, audit trail, fail-closed behaviour, tests, and deployed version have been verified.